# Tarun Ramakrishnan (Tarun R)

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## Research

My research lies at the intersection of **3D computer vision** and **robotics**, with a focus on robot perception algorithms that enable reliable autonomous navigation. Currently, I am particularly interested in visual-inertial 3D reconstruction and localization.

## Education

**MS by Research in Computer Science** Aug 2025 – Present

*International Institute of Information Technology (IIIT) Hyderabad, India*

**Advisors:** Prof. Madhava Krishna and Prof. Sourav Garg **GPA:** 9.50/10.00

**Thesis:** Robot Scene Understanding and Long-term Localization (in progress)

**Coursework:** Advanced NLP, Robotics: Dynamics and Control, Robotics: Planning and Navigation

**B.Tech in Mechatronics Engineering** Aug 2020 – Jun 2024

*SRM Institute of Science and Technology, Kattankulathur, India*

**Thesis:** Campus-Scale Autonomous Ackermann-Steering Robot **GPA:** 8.24/10.00

## Publications and Patents

**SILICA: Repurposing Diffusion Priors for Joint Glass Segmentation and Depth Estimation** Project Page | Code | arXiv

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026

***Tarun R**, Anuj Verma, Laksh Nanwani, Sourav Garg, Madhava Krishna*

**CrowdSurfer: Sampling Optimization Augmented with Vector-Quantized Variational AutoEncoder for Dense Crowd Navigation** Project Page | Code | arXiv

IEEE International Conference on Robotics and Automation (ICRA), 2025

*Naman Kumar*, Antareep Singha*, Laksh Nanwani*, Dhruv Potdar, **Tarun R**, Fatemeh Rastgar, Simon Idoko, Arun Kumar Singh, Madhava Krishna* (* equal contribution)

**System and Method for 3D Point Cloud Rectification Using Depth Sensor with Revolved Reflective Optics**

Indian Patent Application No. 202541131469 (filed, 2025)

*Laksh Nanwani, **Tarun R**, Madhava Krishna*

## Experience

**Teaching Assistant** Jul 2026 – Present

*Mobile Robotics (CS7.503), IIIT Hyderabad* Website

- Lectured on coordinate transforms and common computer vision libraries, prepared real-world visualizations for the SLAM lectures, and guided capstone projects.
- Designed coding assignments and mid-semester exam papers, graded the exams, and conducted vivas.

**Graduate Researcher** Aug 2025 – Present

*Robotics Research Center (RRC), IIIT Hyderabad*

- Developing visual-inertial mapping and relocalization that fuses a learned frontend, a scene coordinate regression network as the implicit map, and IMU preintegration in a factor graph.
- Led SILICA (first author, IROS 2026) and built the Mirage18k dataset of real-world glass walls and panes from 38 indoor scenes.
- Assist my advisors with paper reviews and project proposals, and present my research to visitors including Prof. Takeo Kanade, Prof. Andrew Zisserman, and IEEE RAS members.
- Lead the Smart Wheelchair team; exploring 3D scene-graph-based instruction generation for assisted wayfinding.

**Hardware Lab Administrator** Jul 2025 – Present

*Robotics Research Center (RRC), IIIT Hyderabad* Website

- Manage the robotics hardware lab serving 50+ students, including its 3D printers, simulation servers, hardware procurement, and lab visits.
- Maintain CI/CD pipelines for multiple ground robots, migrated them to current ROS releases, and modernized their power systems and PCBs.

**Research Fellow** Jun 2024 – Jul 2025

*Robotics Research Center (RRC), IIIT Hyderabad* Videos | Code

- Built Phase 2 of the Smart Wheelchair Project from the ground up, developing the full 3D LiDAR-based hardware and software stack on an NVIDIA Jetson Orin NX, including sensor mounts and a power PCB.
- Wrote a real-time C++ 3D LiDAR relocalization system in ROS 2 (global branch-and-bound search with coarse-to-fine registration) and MPC/MPPI controllers, dockerized with CI/CD for multiple wheelchairs.
- Deployed the system in shopping malls, the GMR Airport Innovex center at Hyderabad, office buildings, and academic blocks, running it for hundreds of hours.

**Systems and Mechanical Engineer** (Part-time) Jan 2025 – Jul 2025

*The Energy Labs, India* Videos | Code

- Designed several in-house quadrotors from scratch for different payloads and weights, and manufactured them for sale as commercial products.
- Developed ROS 2 system integration for GPS-denied flight using visual-inertial odometry.

**Drivetrain Engineer** Nov 2020 – Jul 2022

*Team 1.618, Formula Student Team, SRM Institute of Science and Technology*

- Led drivetrain engineering for a 30-member hybrid Formula Student car team, and designed and manufactured a planetary coupling and an eccentric chain tensioner.

## Lectures and Academic Service

**Guest Lecturer, Visual and LiDAR SLAM** – *RRC Summer School 2026* May – Jun 2026

**Guest Lecturer, ROS 2 and Capstone Project** – *RRC Summer School 2025* May – Jun 2025

**Reviewing:** Delegate reviewer for IEEE Robotics and Automation Letters (RA-L) 2025 and IEEE International Conference on Robotics and Automation (ICRA) 2027

## Academic Projects

**Traversability-Aware Indoor Navigation** Slides

- Used diffusion foundation model priors for indoor floor segmentation and depth estimation.
- Wrote a Numba-based GPU MPPI controller, soft-constrained to generate trajectories only within the segmented traversable region.

**Transparent Object Grasping with a Modified SILICA** Slides

- Trained SILICA with CLIP conditioning to predict glass objects and glass scenes separately, greatly reducing false positives.
- Used the model's transparent-object segmentation and depth to generate grasps with NVIDIA GraspGen, and wrote an RRT* planner to reach and pick up the object.

**Omnidirectional Vision-LiDAR Calibration and Fusion Framework** Slides

- Wrote a C++ ROS 2 driver and image stitcher for the Insta360 camera. Since the camera does not timestamp its images, designed a synchronization method from scratch to align its output with a Livox MID-360 LiDAR.
- Built a targetless extrinsic calibration pipeline for a LiDAR, five RealSense cameras, and an Insta360 camera using the LightGlue matcher.
- Used this setup to collect multiple hours of multi-modal data in a crowded shopping mall.

**Surfing Scout the Quadruped** Slides

- Designed a 12-servo controller PCB and wrote C firmware with micro-ROS to interface with the ROS 2 stack.
- Wrote a ROS 2 actuator driver, body-to-leg inverse kinematics, and a C++ IMU-based controller that tracks target body motions.

## Awards and Certifications

**Overall Champions, Formula Imperial (7th season), with Team 1.618** Nov 2021

*Imperial Society of Innovative Engineers (ISIE India)* ID: ISIE/VFI2021/W/076

**Certified SolidWorks Associate (CSWA), Mechanical Design** Aug 2021

*Dassault Systèmes* ID: C-CNMJ6H3WEP

## Skills

**Technical:** Python, C/C++, Linux, ROS/ROS 2, PyTorch, GTSAM, JAX, CasADi, OpenCV, Rerun, SolidWorks
**Languages:** English (fluent), Tamil (native), Telugu, Hindi